

How do oil price expectations influence the economy?

Evidence from OPEC announcements

Känzig (2021)

Warsaw Reading Group, March 2026

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Motivation

Oil prices respond to:

- Supply shocks
- Demand shocks
- Expectations about future supply

Questions:

Do expectation shocks have independent macroeconomic effects? How to identify them?

Contribution

- Identification of *oil supply news shocks*
- Use of high-frequency changes in futures around OPEC meetings
- External-instrument SVAR
- Several robust checks (allowing for IRFs, heteroskedasticity)

Institutional setting

- OPEC coordinates oil supply decisions
- Discrete announcements $\Rightarrow$ information releases

Δ Futures prices $\approx$ News about future supply

OPEC influence

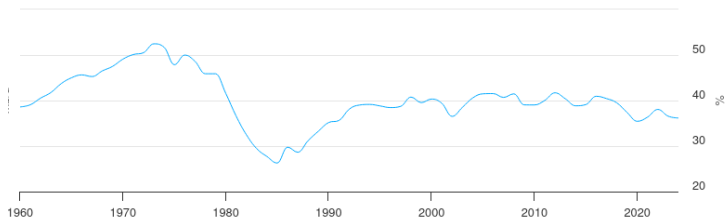


Figure: OPEC share of oil production, OPEC (2026)

High-frequency identification

$$Surprise_m = \alpha_0 + \sum_{j=-1}^2 \theta_j F_{m,yq+j}^{opec} + \sum_{j=-1}^2 \phi_j \left(F_{m,yq+j}^{opec} - F_{m-1,yq+j}^{opec} \right) + IRS_m \quad (1)$$

- $Surprise_m$ – unexpected futures price change in the given month m
- Measures revision in expected oil supply and demand.

VAR representation

$$y_t = b + B_1 y_{t-1} + \cdots + B_p y_{t-p} + u_t \quad (2)$$

$$\text{Var}(u_t) = \Sigma \quad (3)$$

- y_t – vector of endogenous variables:
 - real oil price
 - world oil production
 - world oil inventories
 - world industrial production
 - U.S. industrial production
 - U.S. CPI
- u_t – reduced-form residuals

SVAR model

$$\Sigma = A^{-1}D(A^{-1})' \quad (4)$$

- D diagonal (structural shock variances)

The identification uses oil information shocks as the first variable and treats it as instrument for the identification of matrix $\mathbf{A}$.

Local projections

Using SVAR model makes long-term predictions less volatile, but at the cost of possible errors and misidentifications. The more robust way is to use LPs, which in this work was done as a robustness check.

$$y_{i,t+h} = \beta_{i0} + \psi_{i,h} Shock_t + \beta'_{i,h} x_{t-1} + \xi_{i,t,h} \quad (5)$$

$$LP(h) = \psi_{i,h}$$

Using heteroskedasticity-based identification

Panel A: Heteroskedasticity-based identification

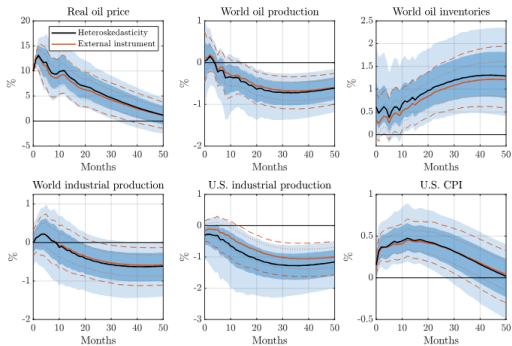


Figure: Känzig (2021)

Using local projections

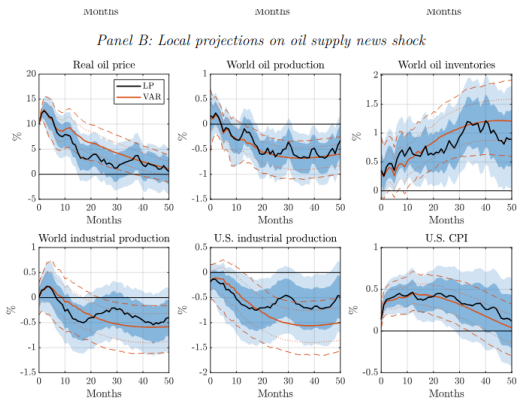


Figure: Känzig (2021)

Oil market responses

- $Shock_t \uparrow \Rightarrow$ oil price $\uparrow$
- $Shock_t \uparrow \Rightarrow$ oil inventories $\uparrow$
- $Shock_t \uparrow \Rightarrow$ oil production $\downarrow$

Historical impact

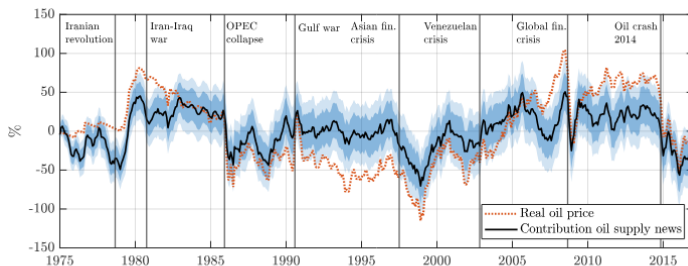


Figure 5: Historical decomposition of the real price of oil

Figure: Känzig (2021)

Channels of influence

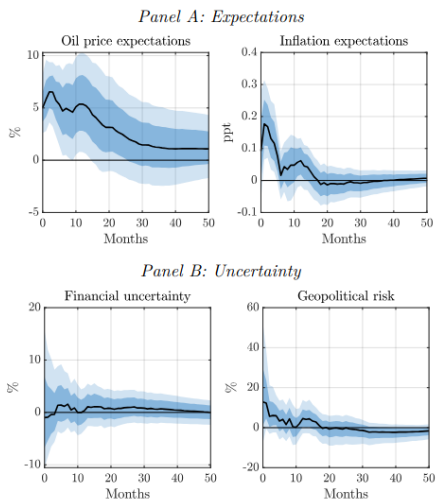


Figure: Känzig (2021)

Comparison to standard supply shocks

$$\varepsilon_t^{news} \neq \varepsilon_t^{supply}$$

- News shocks act before physical changes
- Strong expectations channel
- Different reaction of inventories
- Much stronger reaction of the public than professional forecasters

Takeaways

- Expectations about oil supply matter
- OPEC announcements generate exogenous shocks

Expectations $\Rightarrow$ Prices $\Rightarrow$ Macroeconomy